

5. Electromagnetic (em) waves are used extensively in a wide variety of applications. The table below gives information about different regions of the electromagnetic (em) spectrum. It shows ranges for wavelengths and frequencies.

Region	Wavelength (m)	Frequency (Hz)
radio	$> 1 \times 10^{-1}$	
microwave	1×10^{-3} to 1×10^{-1}	3×10^9 to 3×10^{11}
infra-red		3×10^{11} to 4×10^{14}
visible		4×10^{14} to 7.5×10^{14}
ultraviolet	1×10^{-8} to 4×10^{-7}	7.5×10^{14} to 3×10^{16}
X-ray	1×10^{-11} to 1×10^{-8}	3×10^{16} to 3×10^{19}
gamma ray	$< 1 \times 10^{-11}$	

- (a) Complete the table above using the ranges below. [3]

$(7.5 \times 10^{-7} \text{ to } 1 \times 10^{-3})$ $(< 3 \times 10^9)$ $(> 3 \times 10^{19})$ $(4 \times 10^{-7} \text{ to } 7.5 \times 10^{-7})$

- (b) All em waves transfer energy. The shortest waves are the most ionising. The table below shows energy values in an **incorrect order**. Arrange the energy values in the **correct order**. [1]

Region	Energy (J) in incorrect order	Energy (J) in correct order
radio	$> 2 \times 10^{-14}$	
microwave	3×10^{-19} to 5×10^{-19}	
visible	2×10^{-17} to 2×10^{-14}	
X-ray	2×10^{-24} to 2×10^{-22}	
gamma ray	$< 2 \times 10^{-24}$	

